Mandatory exercise 1

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Question 1

Let $M_{3,3}$ be the vector space of 3×3 matrices, and let A and B be the matrices

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 2 & 1 \\ 2 & 0 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 2 & -2 & 2 \end{bmatrix}$$

Let V be the subset of $M_{3,3}$ given as

$$V = \{C \in M_{3,3} \mid CA = 0\}$$

where 0 is the matrix of all 0's. Is B an element of V? Argue for your answer. Now assume we are given some matrix D in $M_{3,3}$ and form the subset

$$W = \{C \in M_{3,3} \mid CD = 0\}$$

Is W a subspace of $M_{3,3}$? Argue for your answer. Hint: Note that unlike the case of V, I haven't told you what D is. For example, D could be A, and in that case $V = W$, but D could also be any other matrix. If you are unsure of how to approach this problem, try to first check if V is a subspace, then use what you have learned for the more general case of W. In your written solution you should then only include your solution for W.

Is B an element of V?

$$B \in V \iff BA = 0$$

$$BA = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 2 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 3 & 2 & 1 \\ 2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1-3+2 & 2-2+0 & -1-1+2 \\ 0+0+0 & 0+0+0 & 0+0+0 \\ 2-6+4 & 4-4+0 & -2-2+4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Therefore B is an element of V.

Is W a subspace of $M_{3,3}$?

To prove that W is a subspace of $M_{3,3}$, we need to prove the 3 following statements:

1. $W \neq \emptyset$
2. $C_1, C_2 \in W \implies C_1 + C_2 \in W$
3. $C_1 \in W \wedge c \in \mathbb{R} \implies cC_1 \in W$

To prove a matrix C is in W, we need to prove $CD = 0$ for the given matrix D.

Proof:

1. $C_1 = 0 \implies C_1 D = 0 \implies C_1 \in W$
2. $C_1, C_2 \in W \implies C_1 D = 0 \wedge C_2 D = 0 \implies C_1 D + C_2 D = 0$.
   Using the distributive property of matrix multiplication:
   $(C_1 + C_2)D = 0$
3. $C_1 \in W \wedge c \in \mathbb{R} \implies C_1 D = 0 \wedge c \in \mathbb{R} \implies c(C_1 D) = 0$
   Using the associative property of scalar multiplication: $(cC_1)D = 0$

Question 2

For this problem, consider the matrices

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 1 \\ 3 & -1 \end{bmatrix} \qquad C = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$$

Recall that these can be considered as vectors of the vector space $M_{2,2}$ of 2×2 matrices. Are the three matrices A, B, C linearly independent when considered as vectors in $M_{2,2}$? Write out what precisely what it means for them to be linearly independent and argue for your answer.

Determining linear independence:

When removing any element from the set $\{A, B, C\}$ reduces its span, we call the set linearly independent.

The set $\{A, B, C\}$ is linearly independent if:

$$c_1 A + c_2 B + c_3 C = 0 \implies c_1, c_2, c_3 = 0 \text{ is the only solution}$$

To find if this implication is true, we rewrite and solve this equation as a system of linear equations.

$$c_1 \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 3 & -1 \end{bmatrix} + c_3 \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned} c_1 + 2c_3 &= 0 \\ 2c_1 + c_2 + -1c_3 &= 0 \\ 3c_2 + c_3 &= 0 \\ -1c_1 + -1c_2 + c_3 &= 0 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 2 & 1 & -1 & 0 \\ 0 & 3 & 1 & 0 \\ -1 & -1 & 1 & 0 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -5 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & -1 & 3 & 0 \end{array}\right]$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 16 & 0 \\ 0 & 0 & -2 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow \frac{1}{16} R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 0 \end{array}\right]$$

$$\xrightarrow[R_4 \leftarrow R_4 + 2R_3]{\substack{R_1 \leftarrow R_1 - 2R_3 \\ R_2 \leftarrow R_2 + 5R_3}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$c_1 = c_2 = c_3 = 0$$

Using Gauss–Jordan elimination we found that the conclusion of the implication is true. The premise of the implication is given. Therefore the implication must be true and the set $\{A, B, C\}$ is linearly independent.

Question 3

For the matrix A below, compute the following: The rank, the dimension of the row space, the dimensions of the column space, a basis for the nullspace (not just the dimension of the nullspace):

$$A = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 4 & 6 & 4 & 8 & 9 \\ 3 & 4 & 1 & 4 & 8 \end{bmatrix}$$

Finding the Rank, dimension of row space and dimension of column space:

Let's first find the reduced row echelon form of A.

$$A = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 4 & 6 & 4 & 8 & 9 \\ 3 & 4 & 1 & 4 & 8 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 4R_1} \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 0 & -2 & -8 & -4 & 1 \\ 0 & -2 & -8 & -5 & 2 \end{bmatrix}$$

$$\xrightarrow[R_3 \leftarrow R_3 - R_2]{R_1 \leftarrow R_1 + R_2} \begin{bmatrix} 1 & 0 & -5 & -1 & 3 \\ 0 & -2 & -8 & -4 & 1 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow[R_2 \leftarrow R_2 - 4R_3]{R_1 \leftarrow R_1 - R_3} \begin{bmatrix} 1 & 0 & -5 & 0 & 2 \\ 0 & -2 & -8 & 0 & -3 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow[R_3 \leftarrow -R_3]{R_2 \leftarrow -\frac{1}{2}R_2} \begin{bmatrix} 1 & 0 & -5 & 0 & 2 \\ 0 & 1 & 4 & 0 & \frac{3}{2} \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\begin{aligned} Rank(A) &= \text{number of leading 1s in echelon form of } A \\ &= 3 \end{aligned}$$

Theorem: $Rank(A) = dim(R(A)) = dim(C(A))$, where $R(A)$ = row space of A and $C(A)$ = column space of A.
$dim(R(A)) = dim(C(A)) = 3$

Finding a basis for the null space:

For matrix A the null space is defined as:

$$N(A) = \{\overrightarrow{x} \in \mathbb{R}^5 \mid A\overrightarrow{x} = \overrightarrow{0}\} \subseteq \mathbb{R}^5$$

$$\overrightarrow{x} = \begin{bmatrix} v \\ w \\ x \\ y \\ z \end{bmatrix}$$

Because the right side is $\overrightarrow{0}$, we can use the reduced row echelon form we already calculated.

$$\left[\begin{array}{ccccc|c} 1 & 0 & -5 & 0 & 2 & 0 \\ 0 & 1 & 4 & 0 & \frac{3}{2} & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 \end{array}\right]$$

$$
\begin{aligned}
v &= 5s - 2t \\
w &= -4s - \frac{3}{2}t \\
x &= s \\
y &= t \\
z &= t
\end{aligned}
$$

To get the basis vectors we take:
$s, t = 1, 0$ and
$s, t = 0, 1$

$$
u_1 = \begin{bmatrix} 5 \\ -4 \\ 1 \\ 0 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} -2 \\ -\frac{3}{2} \\ 0 \\ 1 \\ 1 \end{bmatrix}
$$